

VALVE POWER BOARD

RUBY - REV A



RESPONSIBLE ENGINEER:
NICHOLAS ALVES

REVIEWERS:

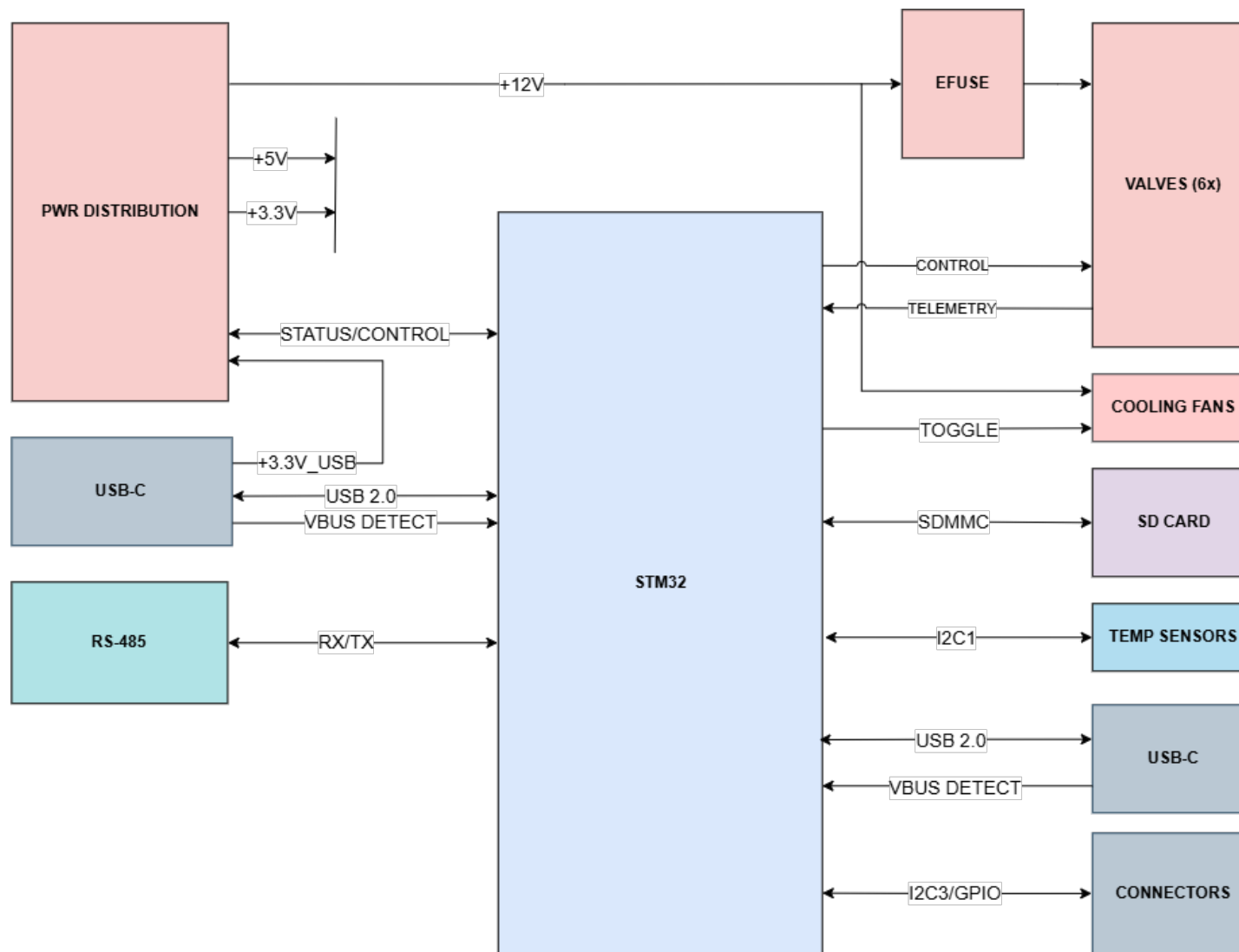
SCRT
STUDENT COMPETITION ROCKETRY TEAM

Sheet: /Title Page/
File: top-level.kicad_sch

Title: TITLE PAGE

Size: A4 Date: 2026-08-27
KiCad E.D.A. 10.0.5

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Sheet: /Block Diagrams/
File: block_diagrams.kicad_sch

Title: BLOCK DIAGRAM

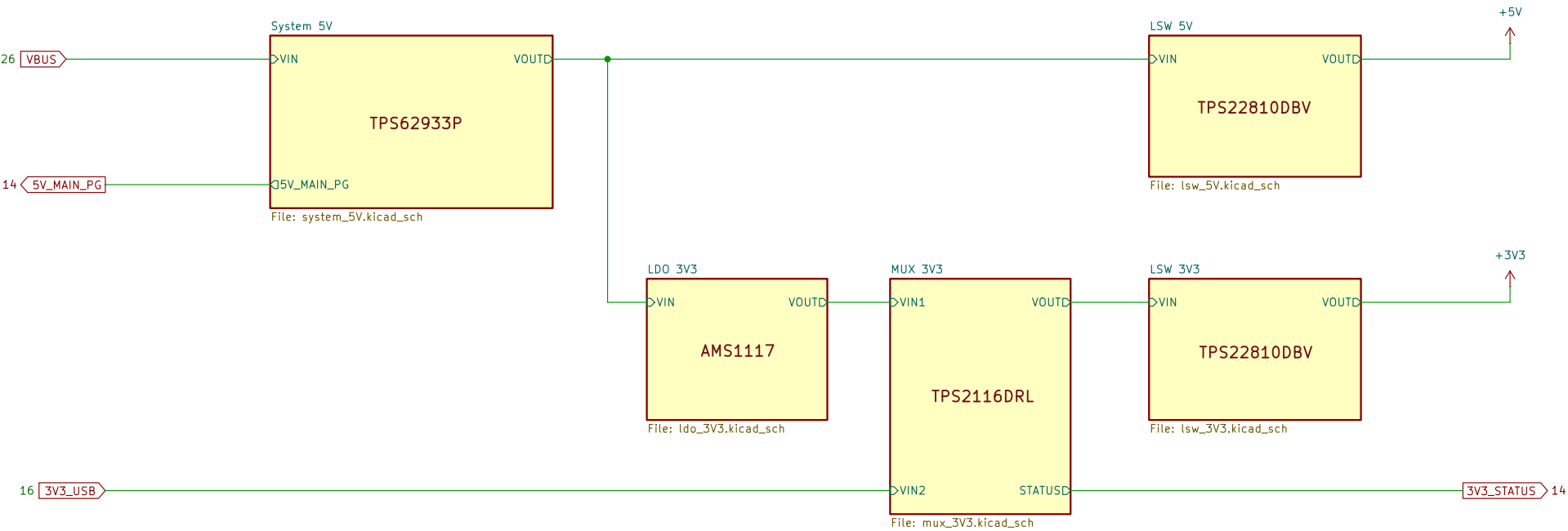
Size: A4 Date: 2026-08-27

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SYSTEM POWER



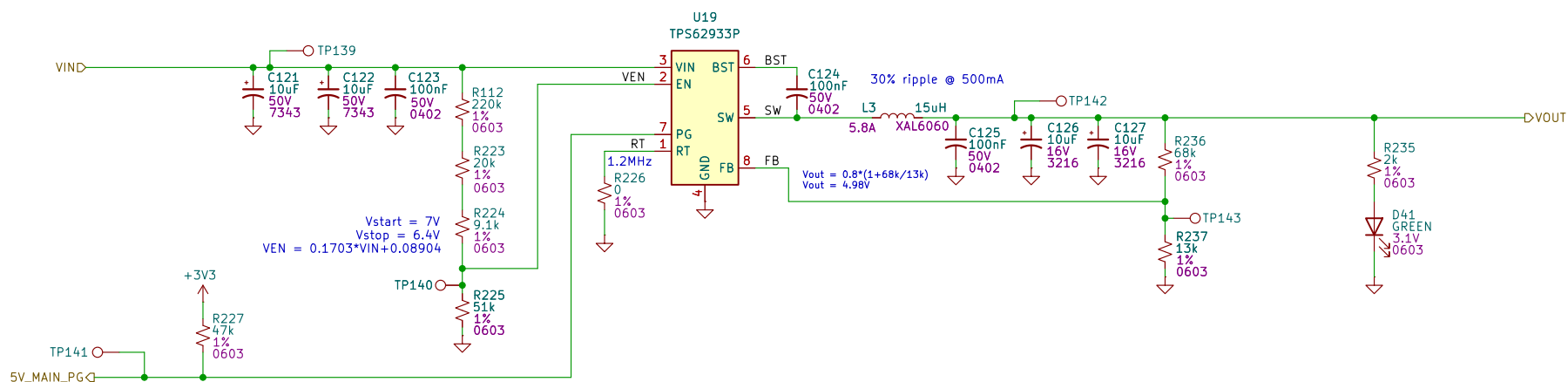


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Sheet: /System Power/
File: system_power.kicad_sch

Title: SYSTEM POWER

Size: A4	Date: 2026-08-27	Rev: A
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Sheet: /System Power/System 5V/
File: system_5V.kicad_sch

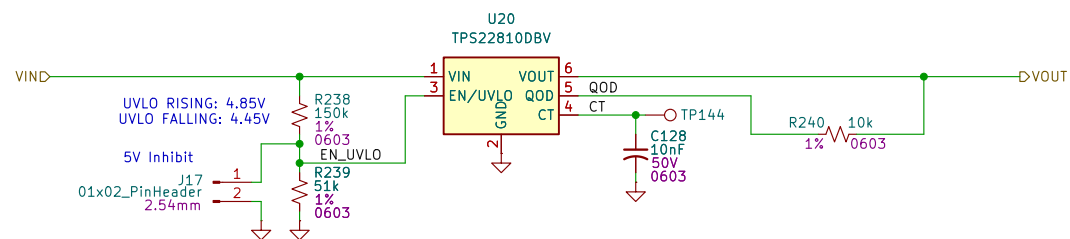
Title: SYSTEM 5V

Size: A4 Date: 2026-08-27

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Sheet: /System Power/LSW 5V/
File: lsw_5V.kicad_sch

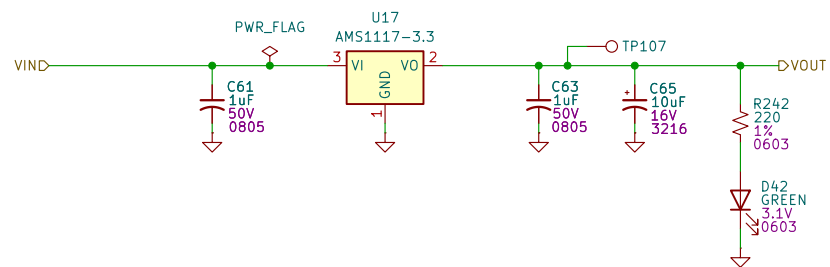
Title: LSW 5V

Size: A4 Date: 2026-08-27

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Sheet: /System Power/LDO 3V3/
File: ldo_3V3.kicad_sch

Title: LDO 3V3

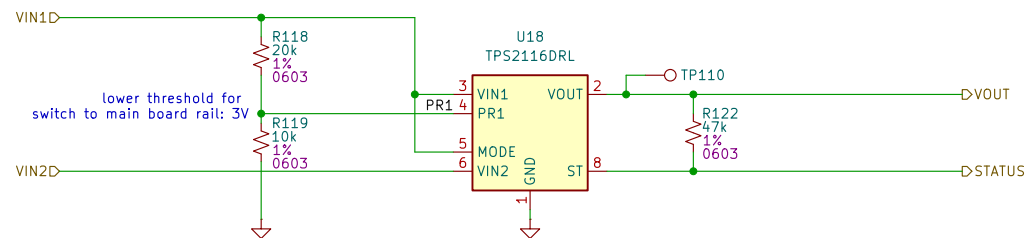
Size: A4

Date: 2026-08-27

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Sheet: /System Power/MUX 3V3/
File: mux_3V3.kicad_sch

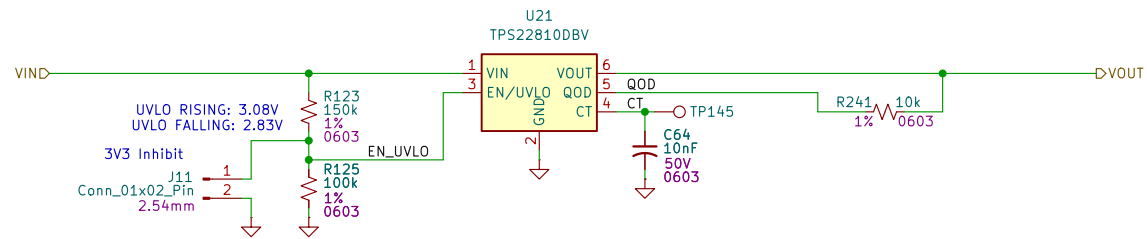
Title: MUX 3V3

Size: A4 Date: 2026-08-27

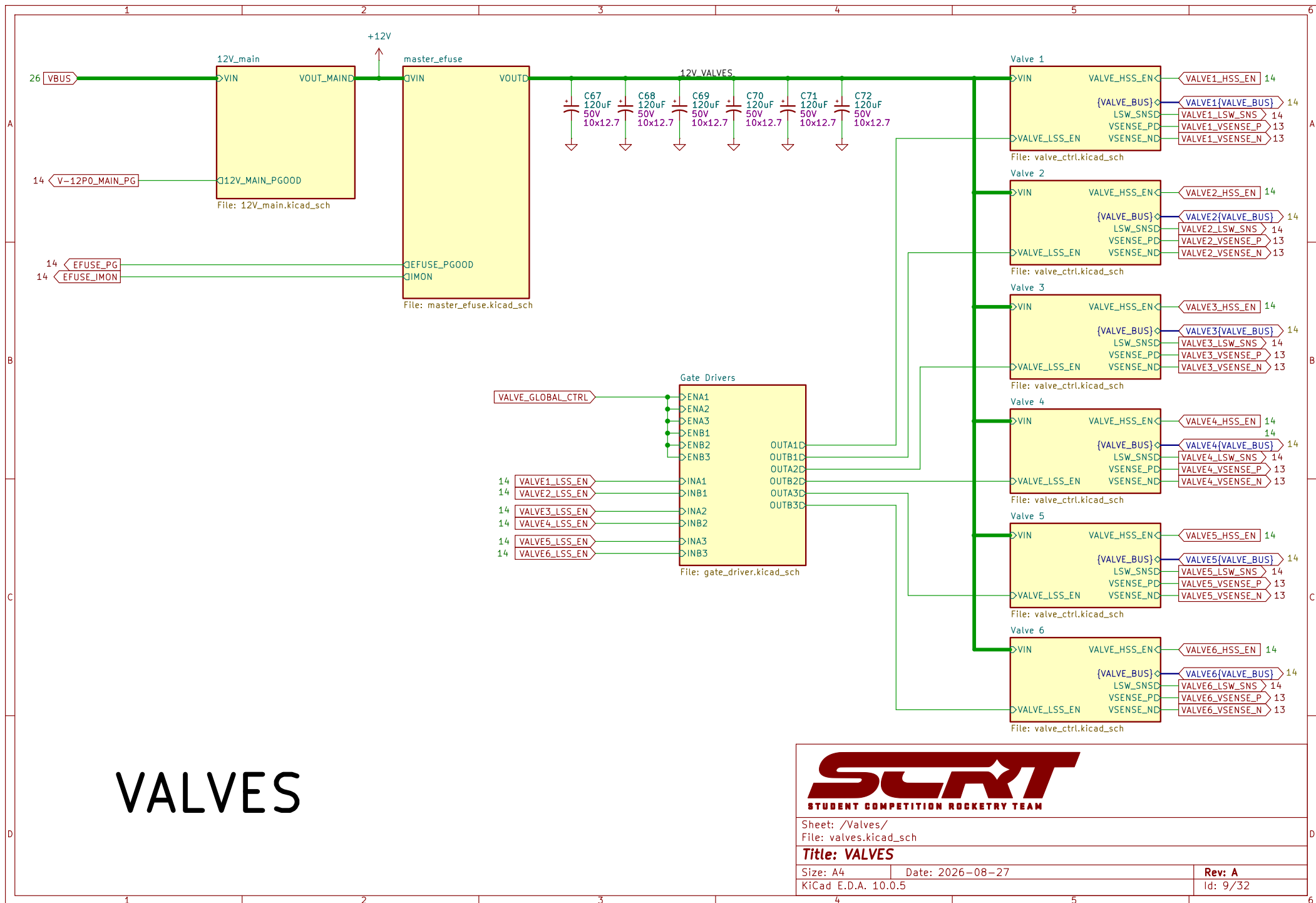
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Sheet: /System Power/LSW 3V3/ File: lsw_3V3.kicad_sch		
Title: LSW 3V3		
Size: A4	Date: 2026-08-27	Rev: A
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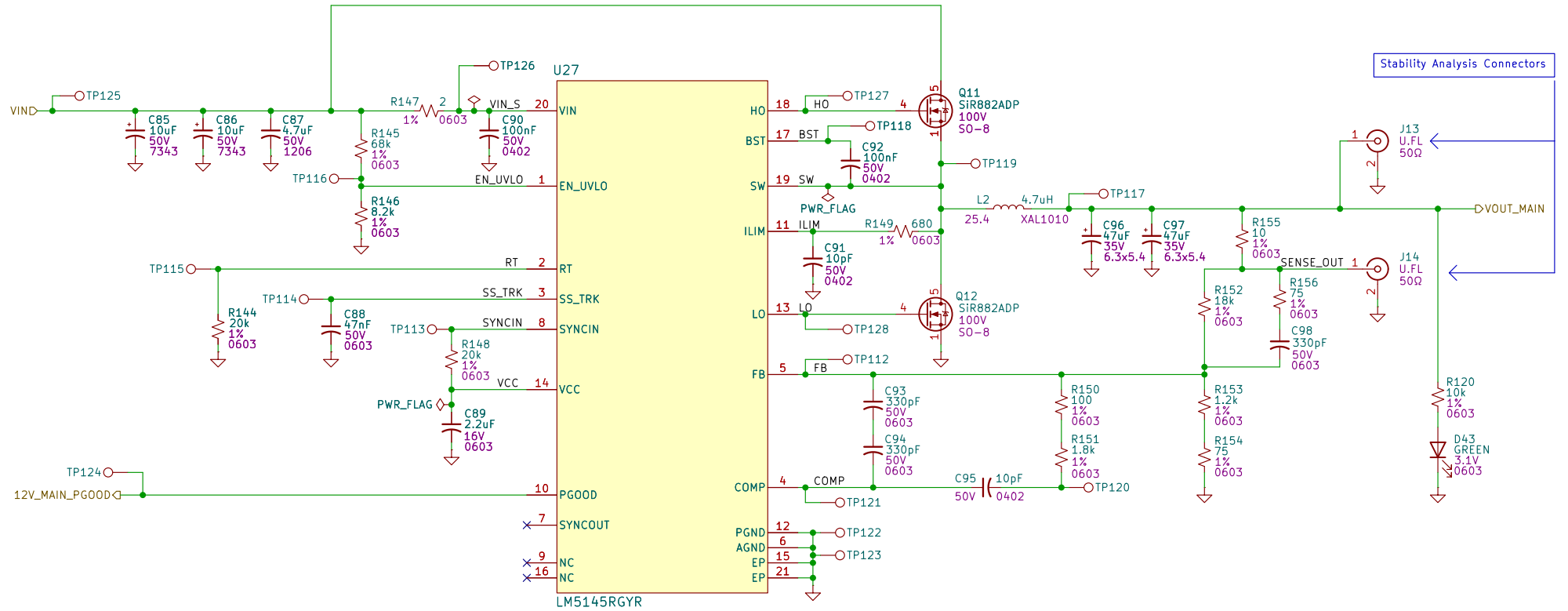


VALVES



Sheet: /Valves/ File: valves.kicad_sch		
Title: VALVES		
Size: A4	Date: 2026-08-27	Rev: A
KiCad E.D.A. 10.0.5		Id: 9/32

12V MAIN



Sheet: /Valves/12V_main/
File: 12V_main.kicad_sch

Title: 12V MAIN

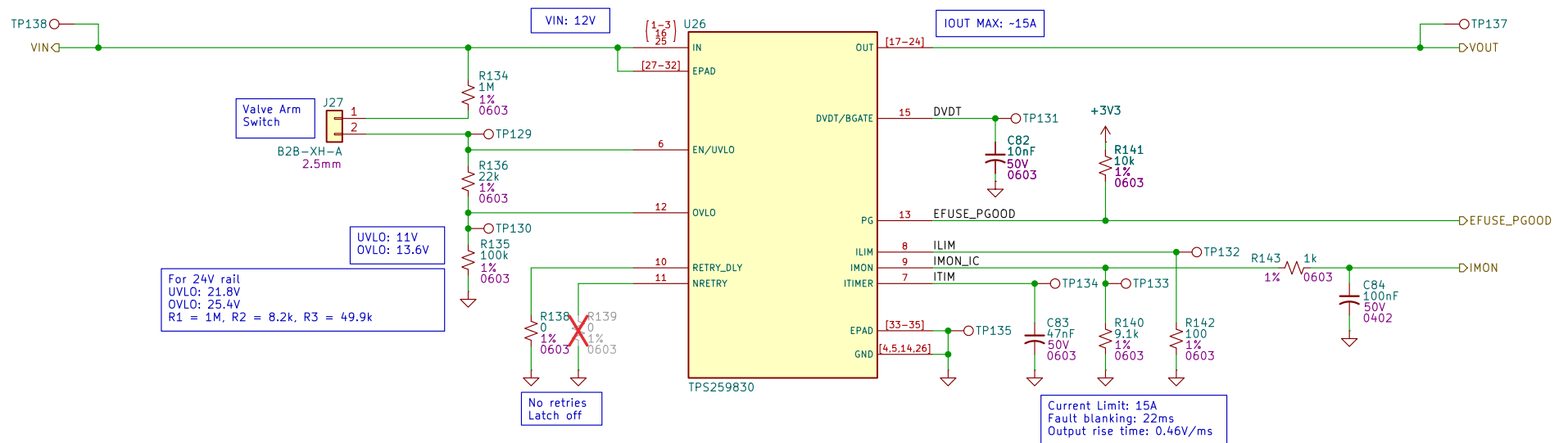
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KiCad E.D.A. 10.0.5

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EFUSE



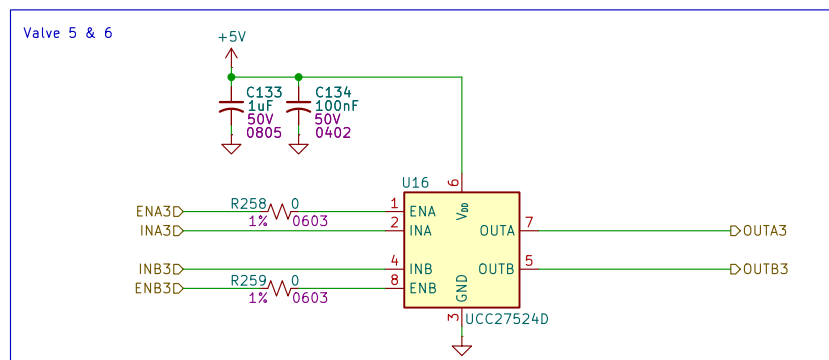
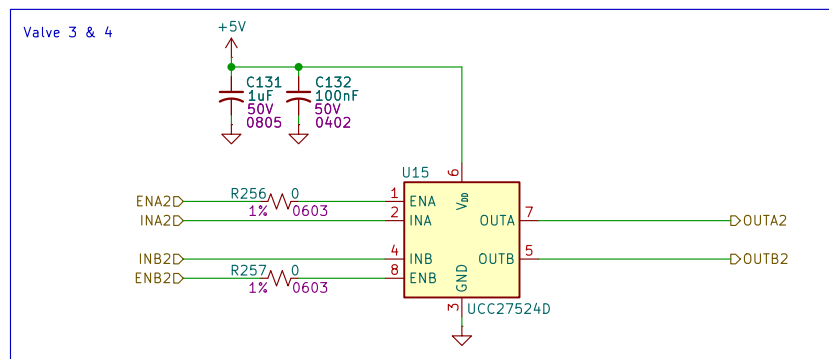
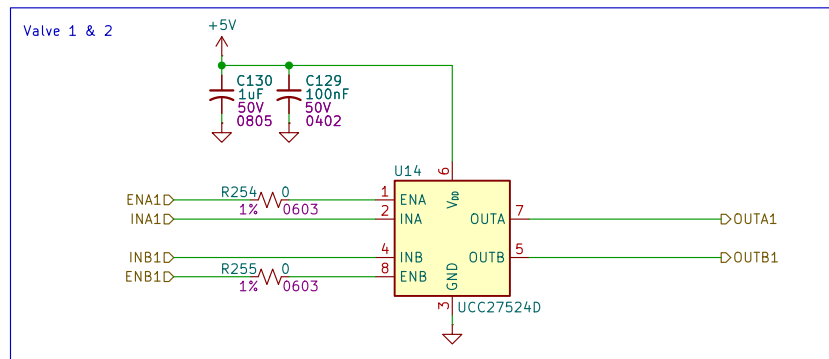


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Sheet: /Valves/master_efuse/
File: master_efuse.kicad_sch

Title: EFUSE

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Sheet: /Valves/Gate Drivers/
File: gate_driver.kicad_sch

Title: Gate Drivers

Size: A4	Date: 2026-08-27	Rev: A
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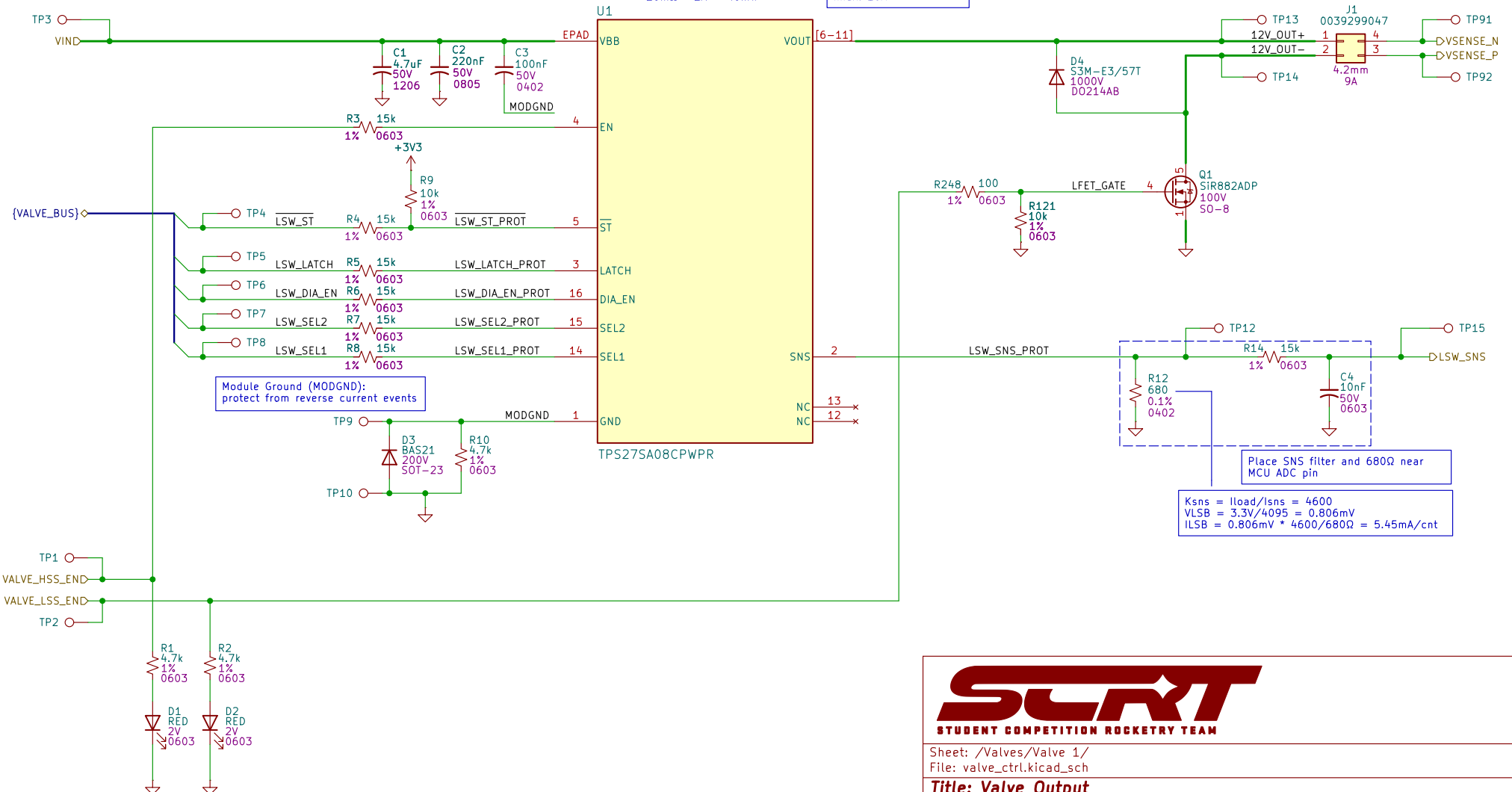
VALVE OUTPUT POWER

Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)

20mΩ * 2A = 40mW

12V, 2A nominal output
Vmax: 36V
Imax: 10A



Sheet: /Valves/Valve 1/
File: valve_ctrl.kicad_sch

Title: Valve Output

Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

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Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)



Title: Valve Output

Title: Valve Output

Size: A4	Date:
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KiCad E.D.A. 10.0.5

Date: 2026-08-27

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Id: 14/32

Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)



File: valve_ctrl.kicad_sch

Title: Valve Output

Size: A4	Date: 2026-08-27
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KiCad E.D.A. 10.0.5

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Id: 15/32

Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)



Title: Valve Output

Title: Valve Output

Title: Valve Output

Size: A4	Date:
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Size: A4	Date:
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KiCad E.D.A. 10.0.5

Date: 2026-08-27

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Id: 16/32

Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)



Title: Valve Output

Id: 17/32

Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)



Title: Valve Output

Title: Valve Output

Title: Valve Output

Size: A4	Date:
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Size: A4	Date:
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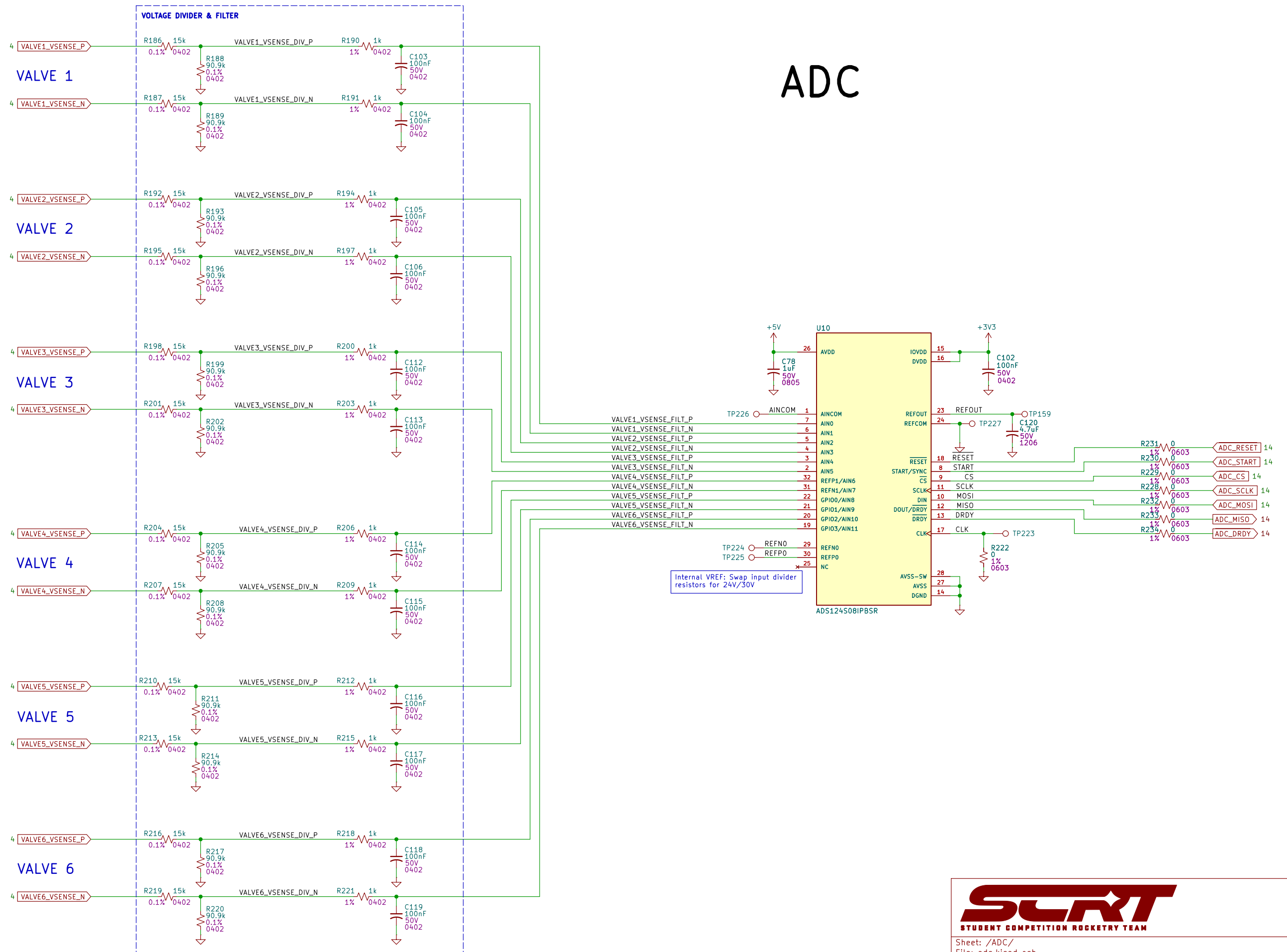
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Date: 2026-08-27

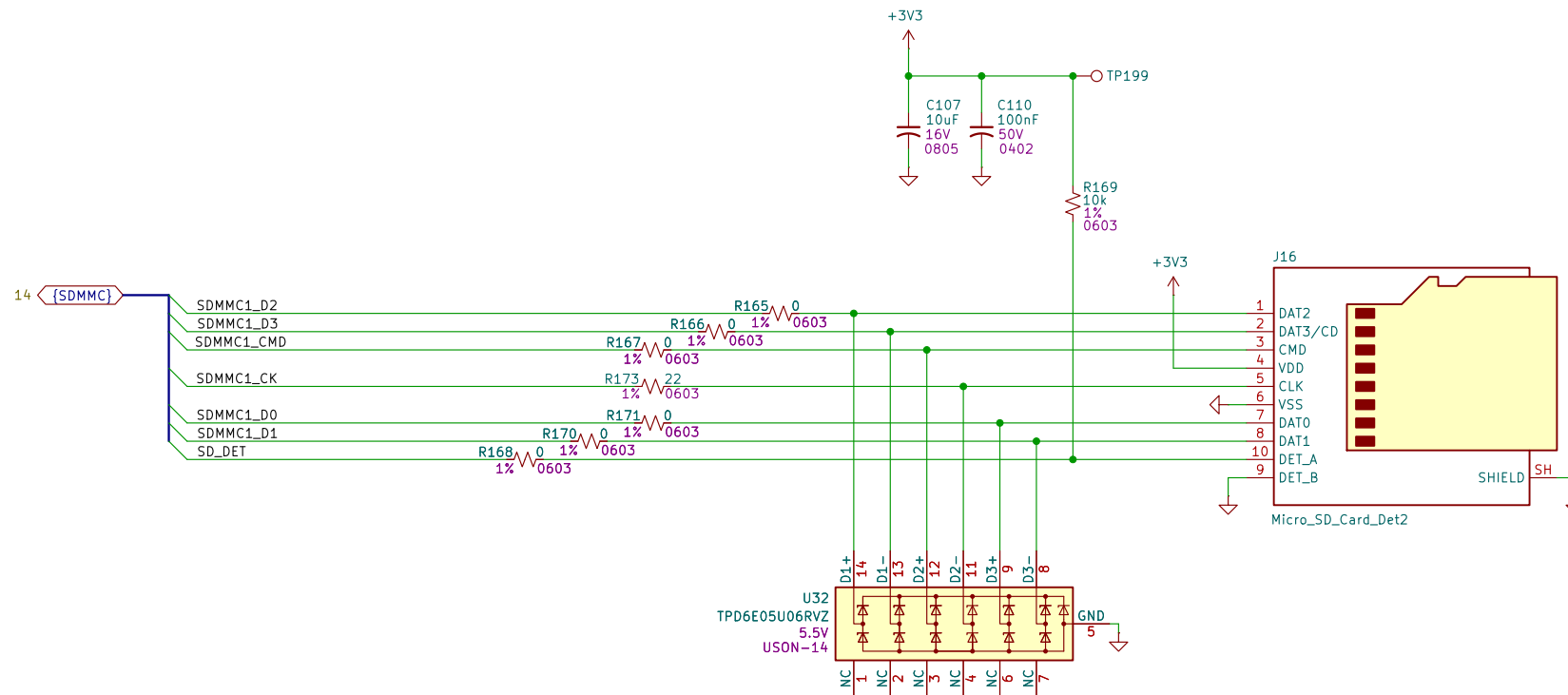
Rev: A

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ADC



SD CARD



Sheet: /SD Card/
File: sd_card.kicad_sch

Title: SD CARD

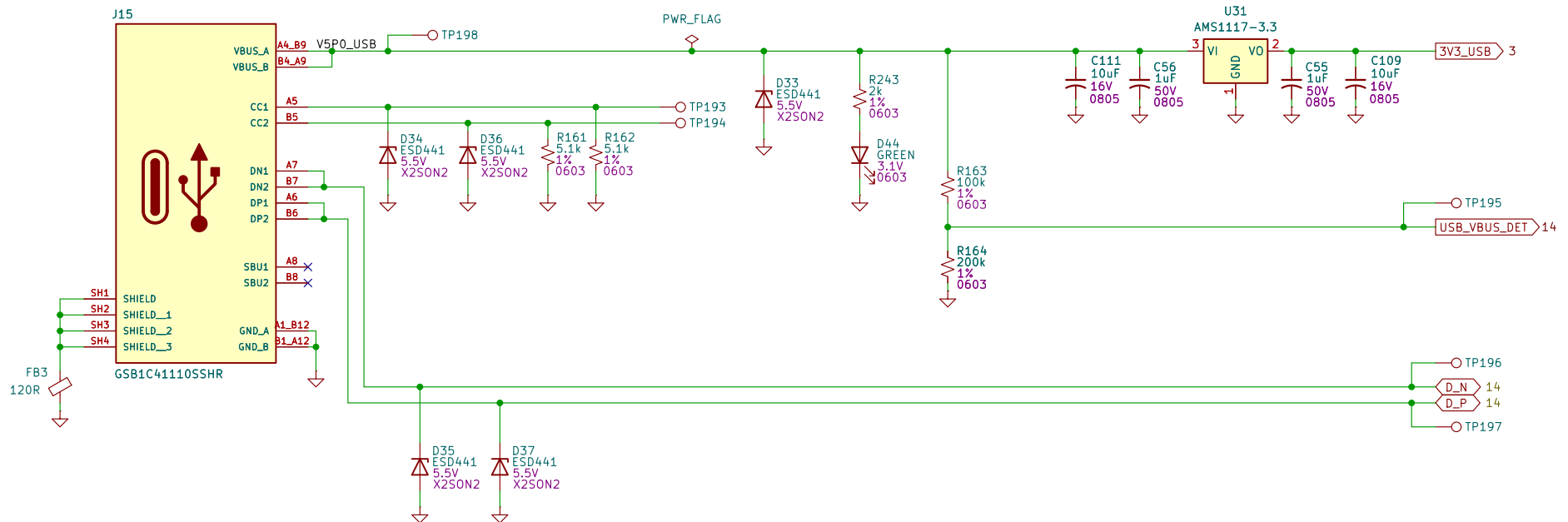
Size: A4	Date: 2026-08-27
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KiCad E.D.A. 10.0.5

Rev: A

Id: 21/32

USB-C



Sheet: /USB-C/
File: usb-c.kicad_sch

Title:

Size: A4

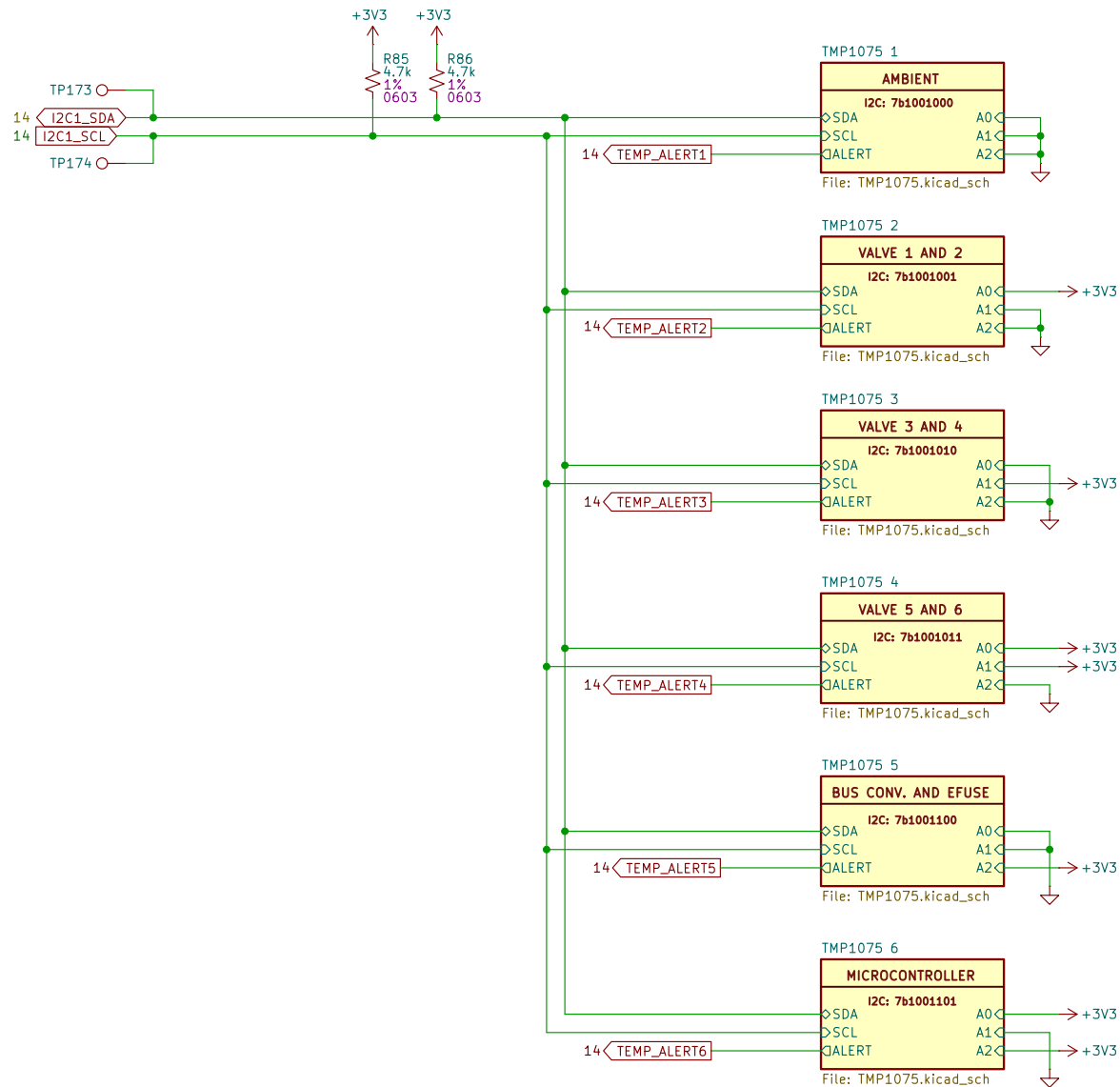
Date: 2026-08-27

Rev: A

KiCad E.D.A. 10.0.5

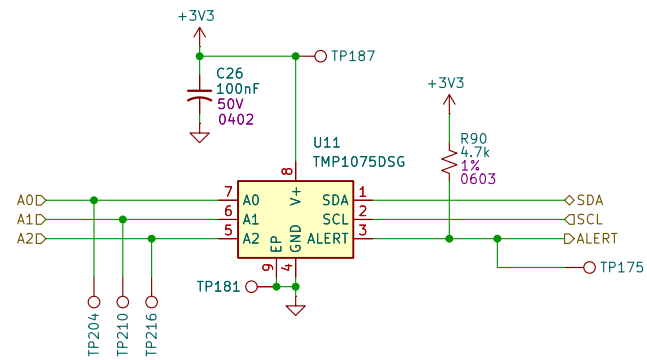
Id: 22/32

TEMPERATURE SENSORS



 STUDENT COMPETITION ROCKETRY TEAM		
Sheet: /Sensors/ File: sensors.kicad_sch		
Title: SENSORS		
Size: A4	Date: 2026-08-27	Rev: A
KiCad E.D.A. 10.0.5		Id: 24/32

TMP1075



Sheet: /Sensors/TMP1075 1/
File: TMP1075.kicad_sch

Title: TMP1075

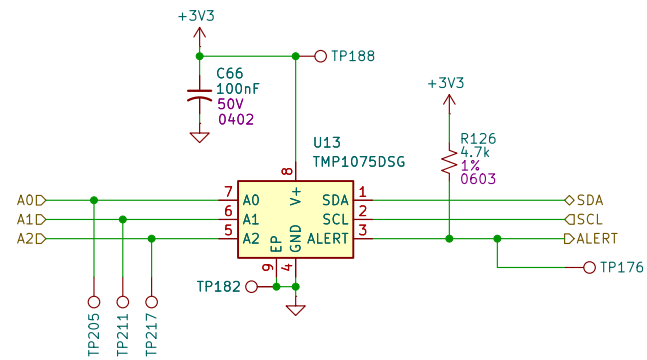
Size: A4 Date: 2026-08-27

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Rev: A

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TMP1075

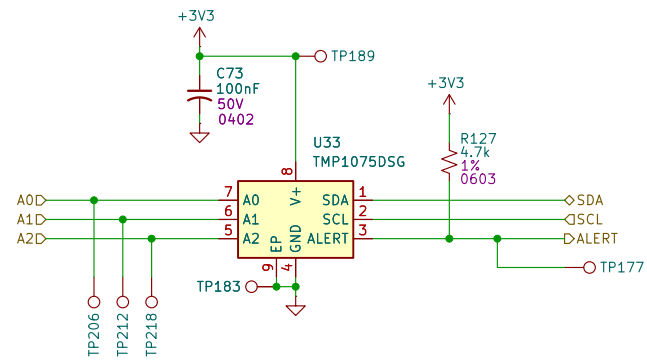


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File: TMP1075.kicad_sch

Title: TMP1075

Size: A4	Date: 2026-08-27	Rev: A
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TMP1075



Sheet: /Sensors/TMP1075 3/
File: TMP1075.kicad_sch

Title: TMP1075

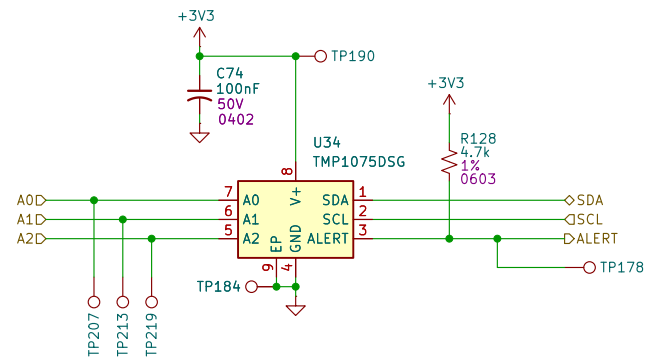
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KiCad E.D.A. 10.0.5

Rev: A

Id: 27/32

TMP1075



Sheet: /Sensors/TMP1075 4/
File: TMP1075.kicad_sch

Title: TMP1075

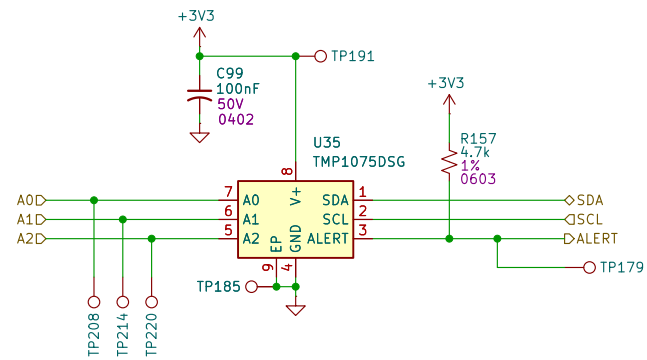
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Id: 28/32

TMP1075



Sheet: /Sensors/TMP1075 5/
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Title: TMP1075

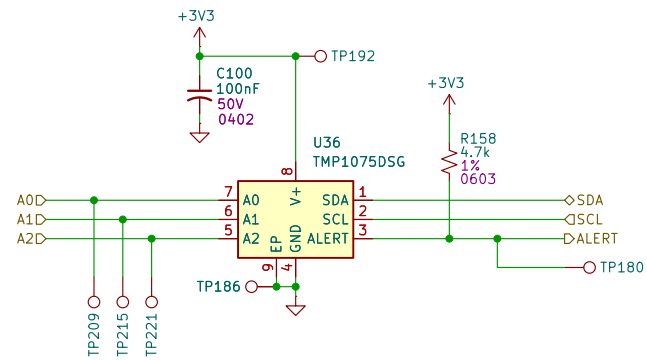
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KiCad E.D.A. 10.0.5

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TMP1075



Sheet: /Sensors/TMP1075 6/
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Title: TMP1075

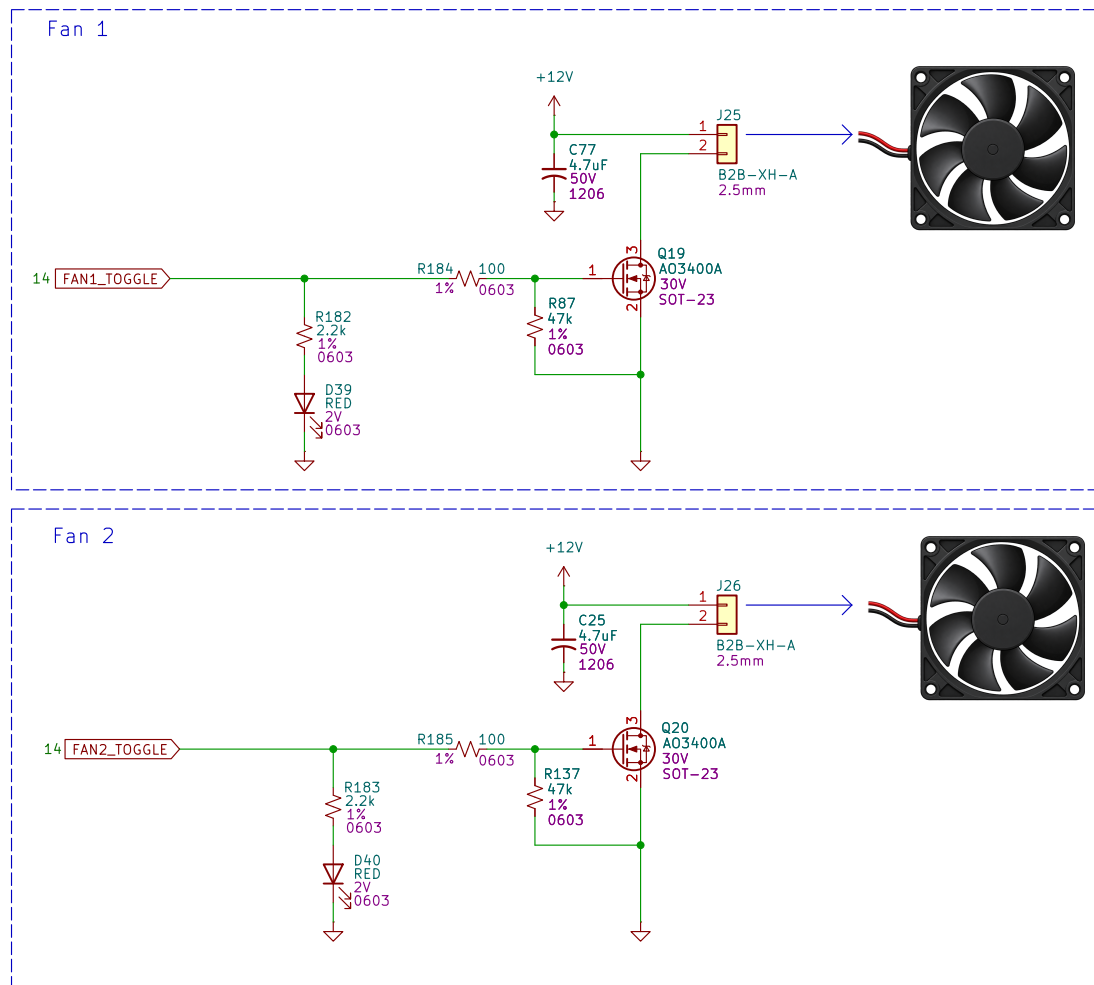
Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 30/32

FANS



Sheet: /Fans/
File: fans.kicad_sch

Title: Fans

Size: A4

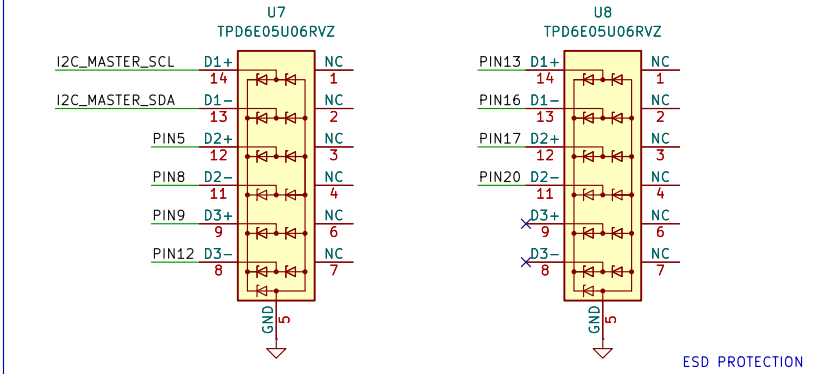
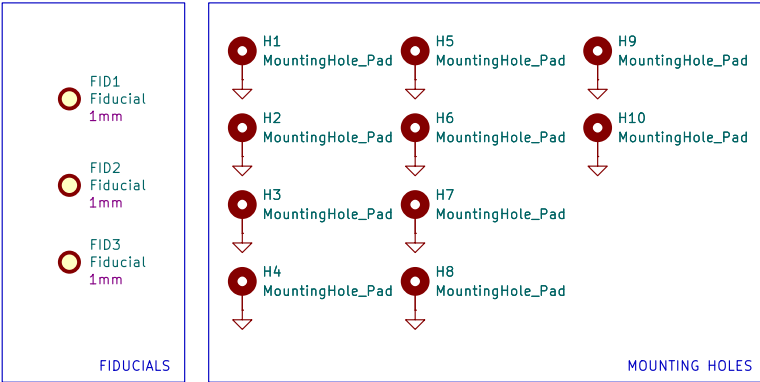
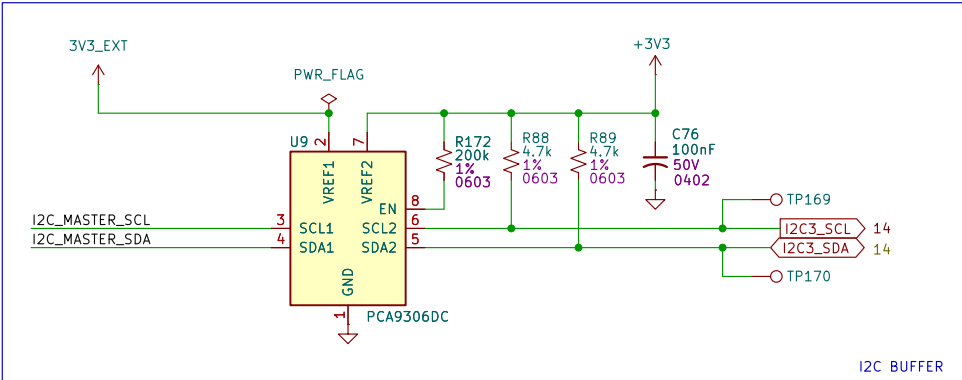
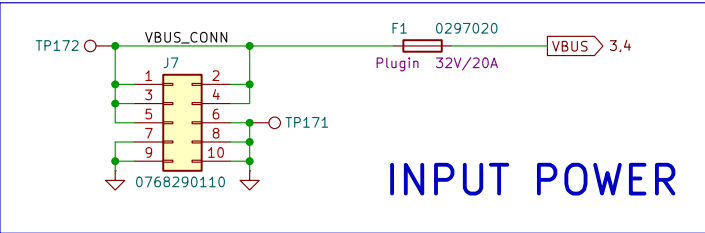
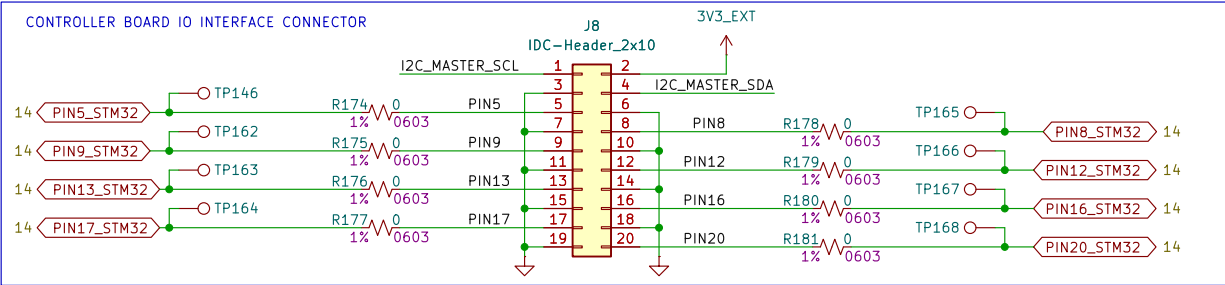
Date: 2026-08-27

KiCad E.D.A. 10.0.5

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MECHANICAL



Sheet: /Connectors/
File: connectors.kicad_sch

Title: MECHANICAL

Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A
Id: 32/32